



Department of Information Technology (Bsc.IT Semester III)

Python Programming Practical

Name: Mehek Sawant	Batch: 2
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Class: SYIT	Date of submission: 14/07/26

Q. 1. Write algorithm for the following:

a. Write a program to perform slicing, basic and advanced indexing on NumPy arrays.

Code:

```
import numpy as np
```

```
#create a 1D NumPy array
```

```
arr1= np.array([10,20,30,40,50,60,70])
```

```
#create a 2D NumPy array
```

```
arr2=np.array([[10,20,30], [40,50,60], [70,80,90] ])
```

```
print("Original 1D Array= ", arr1)
```

```
print("Original 2D Array= ", arr2)
```

```
#Basic Indexing
```

```
print("-----Basic Indexing-----")
```

```
print("Element at index2= ", arr1[2])
```

```
print("Last Element= ", arr1[-1])
```

```
print("Elements at (1,2) of 2D array= ",arr2[1,2])
```

```

#Slicing
print("\n----Slicing----")
print("elements from index 1 to 4= ", arr1[1:5])
print("Every second element= ", arr1[::2])
print("Reversed array= ", arr1[::-1])

#Slicing for 2D(rows 0 to 1, all columns)
print("Sliced(rows 0-1)=\n ", arr2[0:2, :])
#slicing (all rows, col1)
print("Sliced(col1)= ", arr2[:,1])

#Advanced Indexing(Fancy indexing)
print("\n-----Advanced Indexing-----")
indices= [0,2,5]
print("Elements at indices 0,2,5= ", arr1[indices])

#Advanced Indexing for 2D
rows=[0,2]
cols=[1,2]
print("Advanced Indexing(0,2)and(1,2)= ", arr2[rows,cols])

```

OUTPUT:

```

= RESTART: C:/Users/DELL/OneDrive/Desktop/Me
rac4.py
Original 1D Array=  [10 20 30 40 50 60 70]
Original 2D Array=  [[10 20 30]
 [40 50 60]
 [70 80 90]]
-----Basic Indexing-----
Element at index2=  30
Last Element=  70
Elements at (1,2) of 2D array=  60

----Slicing----
elements from index 1 to 4=  [20 30 40 50]
Every second element=  [10 30 50 70]
Reversed array=  [70 60 50 40 30 20 10]
Sliced(rows 0-1)=
 [[10 20 30]
 [40 50 60]]
Sliced(col1)=  [20 50 80]

-----Advanced Indexing-----
Elements at indicies 0,2,5=  [10 30 60]
Advanced Indexing(0,2)and(1,2)=  [20 90]

```

B) Write a program to analyse dimensions and attributes of arrays.

Code:

```
import numpy as np
```

```
# Create 2D Array
```

```
arr = np.array([[1, 2, 3], [4, 5, 6]])
```

```
# Array attributes
```

```
print("Array:\n", arr)
```

```
print("Shape:", arr.shape)    # (rows, columns)
print("Dimensions:", arr.ndim) # No. of dimensions
print("Size:", arr.size)      # Total elements
print("Data Type:", arr.dtype) # Type of elements
print("Item size:", arr.itemsize) # Bytes per element
print("Total Bytes:", arr.nbytes) # Total memory used
```

Output:

```
= RESTART: C:/Users/DELI
rac4.py
Array:
[[1 2 3]
 [4 5 6]]
Shape: (2, 3)
Dimensions: 2
Size: 6
Data Type: int64
Item size: 8
Total Bytes: 48
>>
```